Displayed Order-Book Imbalance and Short-Horizon Midprice Response in Kraken BTC/USD

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Research working paper — exploratory; not peer reviewed or independently replicated.

Abstract

This study examines whether contemporaneous displayed limit-order-book depth imbalance is associated with subsequent Bitcoin/U.S. dollar midprice movements over one to sixty seconds on Kraken. The dataset comprises one nearly 24-hour Level 2 WebSocket recording on 3–4 November 2026, with **12,973,391** recorded book-update messages across **33** separately initialized segments and **86,057** one-second order-book observations. The collection was interrupted by seven reconnect events; their partial files were recovered, and a previous validation run reported exchange CRC32 checks on all segments. A single **1.640-second** local receive-clock reversal was isolated and guarded against in return matching. During this one-day sample, ten-level depth imbalance was positively correlated with five-second forward midprice return (**r** = +**0.3113** pooled), and a within-five-minute correlation of +**0.3136** persisted under retrospective demeaning. A chronological 70/30 imbalance-only model obtained a five-second test $\mathbf{R^2}$ = +**0.0829** against a constant training-mean baseline; a three-feature model achieved +**0.0851** compared with +**0.0093** for momentum plus microprice alone. Train-calibrated imbalance bins in the later test interval showed a +**1.9160 basis-point** top-minus-bottom difference in mean five-second future midprice return. These findings are **within-day exploratory associations**, subject to horizon selection, serial dependence, venue specificity, and development-set reuse. They identify neither causal market impact nor executable trading profitability.

Keywords: Bitcoin; limit-order book; displayed liquidity; depth imbalance; market microstructure; short-horizon forecasting; exploratory time-series analysis

1. Research question and scope

The research question is whether the imbalance of *displayed resting quantities* on the bid and ask sides predicts subsequent midprice changes relative to simple within-day baselines. A separate descriptive component examines bid-ask spreads, visible depth, and the hypothetical cost of walking the currently displayed book.

This distinction matters. Cont, Kukanov and Stoikov (2014) study **order-flow imbalance**, an event-based net supply/demand measure. The predictor here is instead a **stock-of-depth imbalance** from the top ten displayed price levels; the two are not interchangeable. The dataset is a Level 2 *aggregated price-level feed*, not order-level data or a synchronized record of executions, cancellations, and parent orders. Consequently, the research does not identify a causal market-impact response.

2. Data provenance, recording integrity, and eligibility

Venue and feed. Kraken spot BTC/USD, WebSocket v2 `book` channel, depth 100, via `wss://ws.kraken.com/v2`. Kraken describes this channel as aggregated quantities at price levels, with CRC32 checksums of the top ten bid and ask levels (Kraken API Center, 2026). The session ran from approximately **07:31:12 UTC on 3 November** to **07:31:10 UTC on 4 November 2026**. There are 33 segments because the recorder rotated files and reconnected; seven interruptions resulted in `.jsonl.gz.partial` files that were later recovered. The available research scripts require a new snapshot for every segment and do **not** concatenate live book states through reconnects.

Integrity status. The source manifest recorded 10,324,138 updates from the 26 normal-completion files and 2,649,253 updates from seven interrupted segments; after recovery, the combined reported count was **12,973,391**. The terminal validation report supplied with the research chronology stated that all 33 segments passed exchange book CRC32 checks across their update messages, apart from one independent

receive-time warning; the first validator halted before the remainder of segment 5, which a follow-up full pass processed. That exchange validation was *not rerun in the present code-package audit*, because neither raw `.gz` files nor the checksum validator were provided in the audit package. The 33-segment Phase 1 feature processor reported correct update counts and one snapshot per segment. Existing output indicates one backward receive-time jump in segment 5 (11:07:28.223987 to 11:07:26.583757 UTC, approximately 1.640 seconds).

Sampling. At each observed UTC receipt-time second, the feature vector is taken from the **last book state observed within that second**. Seconds without book messages are not interpolated. There are **86,057 segment-level observed seconds**. Labels use the midpoint at **exactly** `t+h` seconds, *within the same segment*, for horizons 1, 5, 10, 30 and 60 seconds. Samples crossing an unobserved second at the required endpoints or flagged clock-anomaly intervals are excluded. The local receive clock may differ from exchange event time, so results are framed as receipt-time observations rather than exchange-timestamp forecasts.

3. Variables and empirical design

Let b_t and a_t denote the best bid and ask, and let $B_t^{(10)}$ and $A_t^{(10)}$ denote cumulative displayed BTC quantities at the best ten levels of each side. The midpoint and imbalance are

$$M_t = \frac{b_t + a_t}{2}, \qquad I_t^{(10)} = \frac{B_t^{(10)} - A_t^{(10)}}{B_t^{(10)} + A_t^{(10)}}.$$

The target is the *future midpoint return*, in basis points:

$$Y_{t,h} = 10^4 \left(\frac{M_{t+h}}{M_t} - 1 \right), \qquad h \in \{1, 5, 10, 30, 60\} \text{ seconds}.$$

The current predictor is fixed using the last received book state in second t; the label is observed only after the later second. **This is not a latency-adjusted executable forecast**: an actual trading system could act only after the book state is received and processed, and would face spread and execution costs.

Additional predictors are last-second midpoint momentum and top-of-book microprice displacement (microprice minus midpoint, in basis points), with microprice computed from the best bid and ask quantities. The Phase 3 linear models use an approximately 70/30 UTC chronological split, hold out the last 30%, and apply a 60-second embargo to test starts. Model parameters and standardization are estimated on the training observations. **Test R^2 is calculated against a constant prediction equal to the training-sample mean future return**; it is not measured against a rich execution or stochastic-price benchmark.

Phase 2 uses pooled correlations, within-segment and retrospectively demeaned five-minute-bucket correlations, nonoverlapping label starts, and per-segment signs. Phase 3 compares imbalance with momentum and microprice and includes 60-second-lagged imbalance and a one-second-delayed response as temporal diagnostics. Phase 4 uses five-minute blocks for bootstrap sensitivity intervals, within-block circular-shift placebos, and ten training-quantile bins evaluated on the later interval. The **five-second primary horizon was selected after inspecting this recording**; all inferential summaries are therefore exploratory.

4. Descriptive order-book conditions

Metric	Full-session mean	Full-session median
Quoted spread (basis points)	0.02162	0.01294
Top-ten displayed bid depth (BTC)	6.3711	5.7073
Top-ten displayed ask depth (BTC)	7.9967	8.1454
Top-ten depth imbalance	-0.10910	-0.16493
Hypothetical instant buy of 0.1 BTC: cost vs midpoint (bps)	0.33219	0.00650

The hypothetical buy estimate walks *currently visible* ask depth and includes crossing the quoted spread, but no trading fees, message latency, hidden liquidity, partial-fill uncertainty, adverse response, or later order-book changes. It is a static book-walk calculation rather than a realized execution result. Depth statistics are one-second-time-weighted over observed seconds, not volume-weighted or trade-weighted.

5. Imbalance and future midprice association

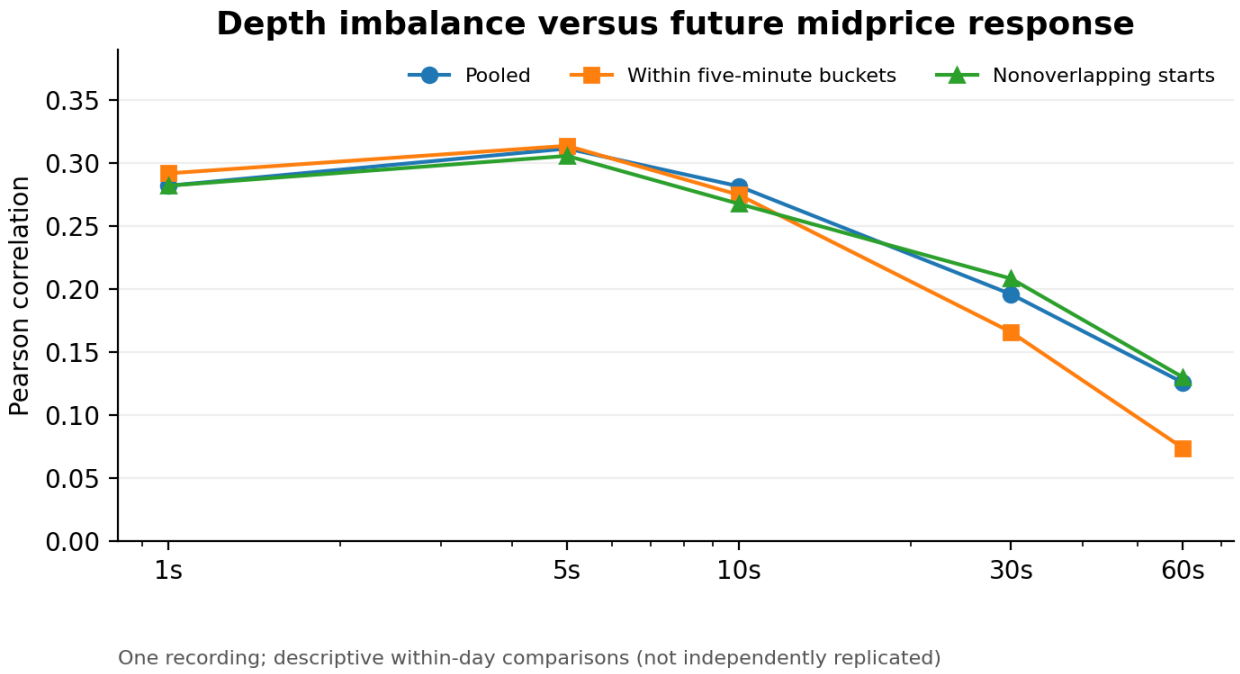


Figure 1: Pooled, within-five-minute, and nonoverlapping correlation by horizon. Each value is computed retrospectively within one recording.

Horizon	Pooled correlation	Within-five-minute correlation	Nonoverlapping correlation	Eligible segment signs
1 s	+0.2819	+0.2917	+0.2819	31/31 positive
5 s	**+0.3113**	**+0.3135**	+0.3055	31/31 positive
10 s	+0.2814	+0.2746	+0.2675	31/31 positive
30 s	+0.1959	+0.1658	+0.2082	31/31 positive
60 s	+0.1259	+0.0740	+0.1301	29/30 positive

For five seconds, 85,663 observations qualified under the Phase 2 rule. The all-day within-five-minute association is positive, as are all 31 segments with at least 100 eligible pairs. These segment observations are not independent day-level experiments: hourly rotations and shorter reconnection segments sample **the same day**, and target returns overlap heavily for 5–60-second horizons.

6. Chronological model comparisons and temporal controls

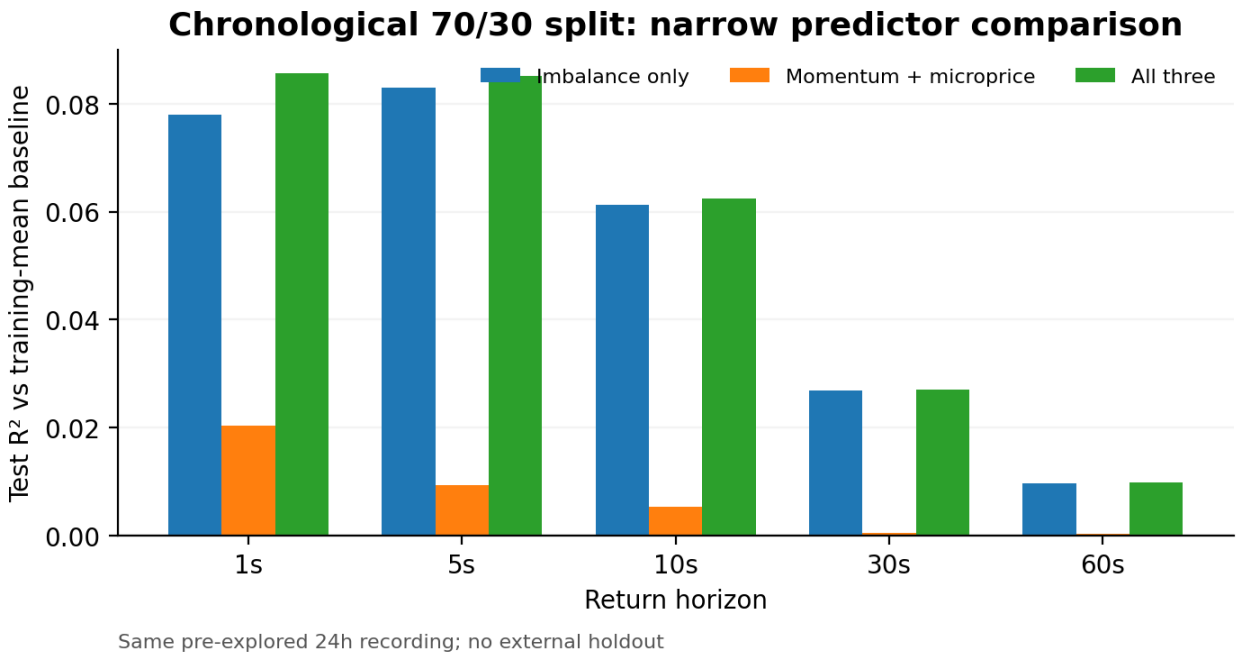


Figure 2: Chronological test R-squared against the constant training-mean forecast. The test period belongs to the same already-examined day.

At five seconds, Phase 3 retained **85,551** pairs that had an observed prior second (needed for the momentum baseline). The later chronological test sample contained **25,698** pairs. The imbalance-only model achieved $\mathbf{R^2}$ = **+0.0829**, momentum plus microprice achieved **+0.0093**, and the three-predictor model achieved **+0.0851**. Thus including imbalance increased test R^2 by **0.0758 points** relative to the narrow momentum-plus-microprice model. This is not a 7.58% trading return. Adding momentum and microprice to imbalance itself produced only a small change in this sample.

A 60-second-old imbalance reading had near-zero correlation with the five-second return on its eligible sample (**-0.0018**), versus **+0.3106** for the current imbalance on those *same* rows. When the response window is delayed by one second, current imbalance remains positively related to the delayed five-second return (**+0.2321**). The tests support short-lived association, but cannot establish a feasible information latency, transaction outcome, or market-impact causality.

7. Block-based stress tests and test-period bins

Using the Phase 4 eligibility protocol, 85,551 five-second pairs were grouped into 295 five-minute blocks. The within-block correlation was **+0.31355**, with a five-minute *block-resampling sensitivity interval* of **[+0.29952, +0.32968]**. A within-block circular-shift placebo (289 eligible blocks, 300 random permutations) produced observed correlation **+0.31406**, placebo 95th-percentile absolute value **0.03593**, and a Monte Carlo two-sided fraction **0.00332**. That fraction is at the resolution floor for 300 random permutations and is **not a formal externally calibrated significance result**. Both exercises assume more independence than the overall single-day evidence can demonstrate.

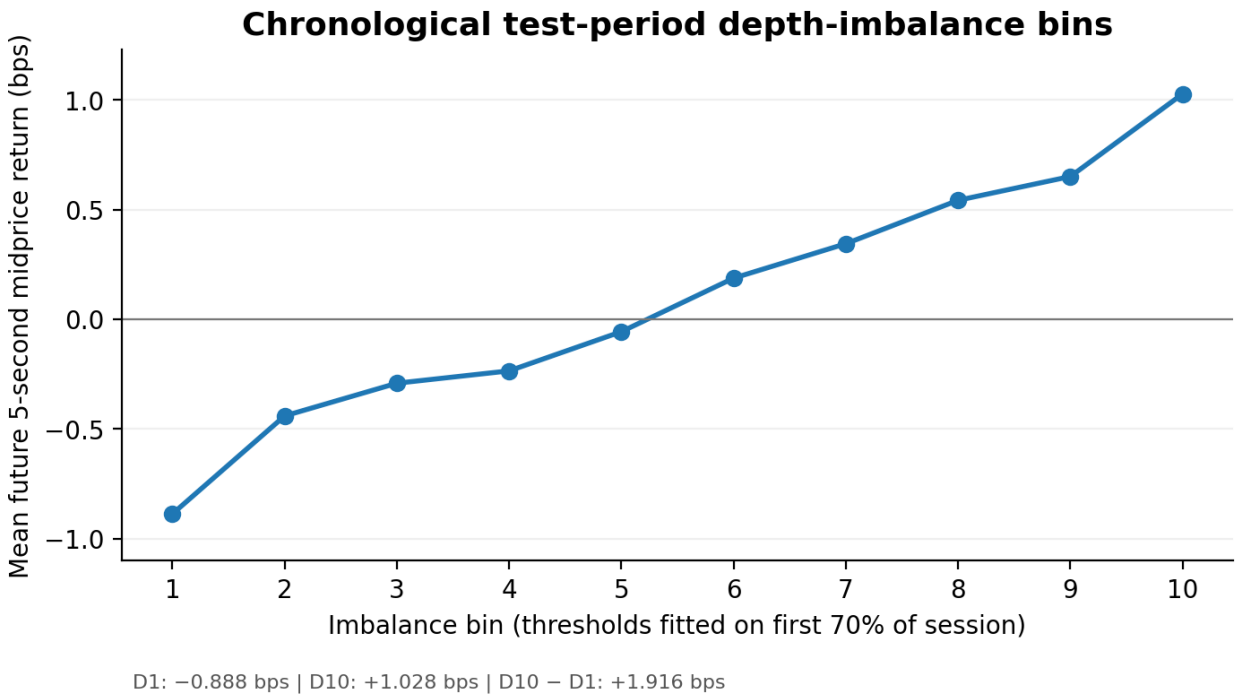


Figure 3: Mean future 5-second midprice return by train-calibrated imbalance bin, evaluated on the later chronological portion of the recording. These are not executable trading profits.

Train-calibrated bin	Test count	Mean 5s future midprice return (bps)
D1	3,284	-0.8877
D2	4,007	-0.4406
D3	2,216	-0.2915
D4	2,246	-0.2352
D5	2,224	-0.0564
D6	2,150	+0.1879
D7	2,242	+0.3450
D8	2,419	+0.5431
D9	2,253	+0.6515
D10	2,660	+1.0283

The test-period D10–D1 difference is **+1.9160 bps**, with a five-minute block-resampling sensitivity interval of **[+1.7414, +2.1113] bps**. The cutoffs were determined from the earlier 70% of the session, so the resulting later-period bin populations are not equal-sized deciles. The near-monotone mean-return profile is a useful descriptive pattern for this day, not a return series from a feasible long–short strategy.

8. Implementation audit and threats to validity

A source-level review of all four analysis scripts, their packaged summaries, and three result CSVs found the following:

1. **Chronology and boundaries.** The scripts derive midpoint targets from exact future-second matches within the same source segment. Phase 3 and Phase 4 require the previous observed second for momentum comparisons and exclude clock-warning intervals. The use of receipt-second last-book features is consistent across phases.

2. **Model fitting.** Phase 3 estimates linear-model centering, scaling, and coefficients from training rows only and uses an embargo between training labels and test starts. The implementation uses a tiny ridge stabilization on predictor diagonals; it is effectively a near-unregularized linear regression, not an entirely unpenalized OLS estimator.
3. **Retrospective adjustments.** Five-minute bucket demeaning uses outcomes from the full bucket and must not be construed as an online predictor. Decile thresholds are training-period-only, though the five-second selection was not preregistered.
4. **Statistical dependence.** Consecutive book states are strongly serially related; adjacent five-minute blocks and overlapping targets can remain dependent. Block resampling quantifies **within-day sensitivity**, not a day-level confidence interval.
5. **Limited comparator set.** Momentum and microprice are useful narrow controls, but the study does not compare against rich order-flow-imbalance measures, stochastic-price predictors, time-of-day interaction models, or other exchanges.
6. **Unreproduced inputs.** This review rechecked Python compilation, numerical-result consistency, and selected synthetic fixture behaviors, but **did not rerun the 12.97-million-message pipeline**. The audit package contains source code, final summaries, and per-segment and decile tables, but **not** the 86,057-row feature file, source manifest, source recordings, or standalone checksum validator. Results are therefore **not independently reproducible from this release package alone**.

9. Conclusion

Within one recording of Kraken BTC/USD displayed limit-order-book depth, ten-level bid/ask imbalance exhibited an unusually clear, short-lived association with future midprice movements. This association remains visible after several informative within-day checks, including fixed-interval nonoverlapping starts, retrospective within-bucket demeaning, a chronological model comparison, temporal lags and delayed response windows, and test-period imbalance bins calibrated from the earlier part of the day. The findings are best understood as an **exploratory microstructure observation**, not a causal identification, peer-reviewed result, independently validated multi-day regularity, or investable alpha claim.

The most valuable next experiment is a **new, untouched day** with the feature definition, five-second horizon, frozen calibration rule, and evaluation protocol specified in advance. An independently run reconstruction and checksum audit from preserved source data would address the remaining software reproducibility gap.

Data, code, and reproducibility

All reported numbers in this manuscript are transcribed from the uploaded package's `phase1_summary.json`, `robustness_summary.json`, `incremental_validation_summary.json`, `phase4_statistical_stress.json`, and associated CSV tables. Four source scripts and archived outputs accompany this draft. Reconstructing observations requires separately preserved Kraken WebSocket segment files and the original source manifest. Re-running the integrity audit also requires the checksum validator. Exchange quotes are not guaranteed executable at quoted size in a live strategy. The manuscript itself does not claim external replication or peer review.

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